

# Supplementary Material for

## “Certifying the Outcome of Autonomous Vehicle Fallback Triggers: Closed-Loop Evaluation Under Latency and Dataset Shift”

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This document contains the exploratory and post hoc analyses summarized in Section V of the main paper. All numbers use the same settings as the main paper ( $\alpha = 0.05$ ,  $\delta = 0.10$ , 1,994 Argoverse 2 calibration and 1,996 test scenarios, 1,504 nuScenes scenarios) and can be regenerated from the released per-scenario records without re-simulation.

### S1. CLASS-CONDITIONAL CERTIFICATION (POST HOC)

Table S1 lists the differences between the closest prior work and this study.

TABLE S1  
LUO *et al.* VERSUS THIS STUDY.

|                   | Luo <i>et al.</i>                                                           | This paper                                                                     |
|-------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Guarantee         | $\Pr[\text{no alert} \mid \text{unsafe}] \leq \epsilon$ , class-conditional | $\Pr[\text{harm not pre-empted}] \leq \alpha$ over scenarios, high probability |
| Unsafe label      | safety score of logged future $< f_0$                                       | nuPlan harm in the no-fallback run                                             |
| Predictor         | Trajectron++                                                                | AutoBot                                                                        |
| Alert consequence | none (scored on logs)                                                       | MRM executed in closed loop                                                    |
| Cost of alerting  | false-positive rate                                                         | unnecessary stops, induced collisions, residual harm                           |
| Data              | nuScenes, Lyft                                                              | Argoverse 2, nuScenes, Waymo                                                   |
| Shift             | label shift only                                                            | dataset shift, weighted repair                                                 |
| Latency           | not studied                                                                 | studied                                                                        |

Table S2 implements the certificate of Luo *et al.* (cited in the main paper) in our framework: Mondrian split conformal prediction on the unsafe class, where unsafe means harm in the reference run and an alert means firing before it, with the finite-sample correction and without their tie randomization. At  $\epsilon = 0.05$  on the 316 unsafe calibration scenarios it selects a threshold equivalent to always firing, so the class-conditional target is vacuous for this predictor at this level. The score separates harmful from harmless scenarios well enough for a marginal target (AUROC 0.83) but not for a 5% miss rate among harmful ones. At the marginal-matched  $\epsilon = \alpha/\pi = 0.32$  it selects the same threshold as empirical tuning. Deployed on nuScenes, that certificate has a conditional miss rate of 66% against its 32% target. The harm base rate rises from 15.8% to 22.1%, but the conditional miss rate is far above what that alone explains, so the class-conditional guarantee does not survive the dataset shift either.

### S2. REVERSE SHIFT (EXPLORATORY, POST HOC)

We also certified on half of the nuScenes scenarios and deployed on Argoverse 2 test, a direction pre-registered as

TABLE S2  
MARGINAL CERTIFICATE (LTT) VERSUS CLASS-CONDITIONAL CERTIFICATES (LUO-STYLE), THRESHOLDS CERTIFIED ON ARGOVERSE 2. MISS: MARGINAL MISS RATE; COND.: MISS RATE AMONG SCENARIOS WITH HARM IN THE REFERENCE RUN. RATES IN %.

| Certificate                   | Argoverse 2 test |       |        | nuScenes |       |        |
|-------------------------------|------------------|-------|--------|----------|-------|--------|
|                               | Miss             | Cond. | Unnec. | Miss     | Cond. | Unnec. |
| LTT, marginal $\alpha = 0.05$ | 3.7              | 24.2  | 28.5   | 13.2     | 58.1  | 26.1   |
| Luo-style, $\epsilon = 0.05$  | 0.0              | 0.0   | 84.9   | 5.0      | 19.6  | 77.9   |
| Luo-style, $\epsilon = 0.32$  | 4.3              | 28.5  | 26.0   | 14.9     | 66.0  | 21.3   |

a reverse-shift analysis. As pre-registered, with all scenarios, the procedure *refused to certify* on both halves, because of the 5.0% floor. On the avoidable subset (post hoc), thresholds certified on nuScenes ( $n \approx 715$ ) give an Argoverse 2 miss rate of 1.9% and 2.0%, both below the target, at 40.6% and 39.5% unnecessary stops. A certificate from the harder dataset therefore transfers safely but conservatively, while the reverse does not transfer. This asymmetry is a single-pair observation pending replication.

### S3. RECALIBRATION WITH A FEW LABELED TARGET SCENES

Table S3 draws  $k$  labeled target scenes at random and evaluates on the rest. The nuScenes analysis (avoidable subset, 60 draws) was exploratory and not pre-registered. Its Waymo counterpart (all scenarios, 200 draws) was pre-registered as R1–R4 before any Waymo outcome was read, and its outcomes are reported in Section V of the main paper. On nuScenes, uncertified tuning on 40–60 target scenes exceeds the target in about two draws in three. A certified threshold on the same scenes keeps the target (violations 0–10%) but stops in 53–81% of scenarios, against 27% without target labels. A handful of labeled scenes therefore restores a *valid* certificate, not the original operating point. On Waymo, uncertified tuning violates the target in 69.5% and 59% of draws with 40 and 60 target scenes, and weighting the Argoverse 2 calibration set with 60 Waymo scenes ( $n_{\text{eff}} = 232$ ) keeps validity at 0% violation.

### S4. SIMULATOR SENSITIVITY

Three changes to the simulator, each paired by scenario seed against the main run (reactive traffic, 4.0 m/s<sup>2</sup>, 2 Hz), leave every conclusion of the main paper unchanged. Non-reactive log-replay traffic raises the harm base rate (17.7%

TABLE S3

RECALIBRATION WITH  $k = 60$  LABELED TARGET SCENES. NUSCENES: EXPLORATORY, AVOIDABLE SUBSET, 60 DRAWS. WAYMO: PRE-REGISTERED (R1–R4), ALL SCENARIOS, 200 DRAWS. VIOL.: FRACTION OF DRAWS WITH TEST MISS  $> \alpha$ . RATES IN %.

| Method                   | nuScenes |       |        | Waymo |       |        |
|--------------------------|----------|-------|--------|-------|-------|--------|
|                          | Miss     | Viol. | Unnec. | Miss  | Viol. | Unnec. |
| AV2-certified, no labels | 9.2      | –     | 27.1   | 2.7   | 0     | 45.0   |
| Target-only tuned        | 6.0      | 70    | 36.7   | 6.2   | 59    | 29.6   |
| Target-only LTT          | 0.8      | 0     | 72.5   | 1.5   | 2.5   | 58.6   |
| AV2 + 60, weighted LTT   | 2.8      | 0     | 53.3   | 1.7   | 0     | 49.8   |

versus the main run’s 15.1 %, since logged agents cannot react to an ego that deviates from its log) and, with it, unnecessary stops by 7.9 points [95 % CI 5.3, 10.8]. The miss rate itself does not move (paired difference  $-0.5$  points  $[-1.3, 0.0]$ ). A  $2.0 \text{ m/s}^2$  comfort deceleration changes neither miss nor stops (paired difference exactly 0 in both), because both alert-level quantities depend only on the reference run and the firing tick (Observation 1 of the main paper), not on how hard the vehicle then brakes. It does lower induced collisions (2.6 % to 1.6 %), the one quantity that does depend on post-fire kinematics. Decisions at 10 Hz on 100 scenarios give a miss rate of 2.9 % [1.0, 8.2], consistent with the main run within an interval widened by the smaller sample.

#### S5. CERTIFYING RESIDUAL HARM DIRECTLY (POST HOC)

*Update.* The attempt described here, with a threshold ordered by the monotone alert-level order and a 5 % target, failed. A later amendment (Amendment 11) certified the same quantity at a 7.5 % target with thresholds ordered by out-of-fold estimated risk, and succeeded (Section V-D of the main paper). The failure below is therefore attributable to the target and the order, not to outcome-level certification as such.

The certificate in the main paper bounds the alert indicator  $L(X, \lambda)$ , not harm in the run the vehicle then drives. We therefore tried to certify *residual harm*  $H(X, \lambda)$ , the indicator that the nuPlan harm definition fires anywhere in the executed run, before or after the fallback engages.  $H$  is not monotone in  $\lambda$ , because firing earlier can itself create harm (4.5 % of fresh calibration scenarios have a non-monotone  $H$ ), so Conformal Risk Control does not apply by construction. Learn-then-Test needs no monotonicity but is tested on the same fresh Argoverse 2 split as the replication analysis of the main paper ( $n_{\text{cal}} = n_{\text{test}} = 904$ , target  $\alpha_H = 0.05$ ,  $\delta = 0.10$ ). The comparisons were pre-registered as N1–N5 before running.

TABLE S4

CERTIFYING RESIDUAL HARM ON THE FRESH SPLIT. RATES IN %, WILSON 95 % INTERVALS. VIOL.: FRACTION OF 200 RESPLITS WHOSE TEST RESIDUAL HARM EXCEEDS 5 %. THE SECOND PREDICTOR GIVES THE SAME PICTURE (OUTCOME-LEVEL LTT: 2.8 % HARM, 84.8 % STOPS, 16.0 % VIOLATIONS).

| Certificate on               | Resid. harm [CI] | Miss | Stops | Viol. |
|------------------------------|------------------|------|-------|-------|
| Alert $L$ , LTT (main paper) | 5.0 [3.7, 6.6]   | 3.4  | 29.0  | –     |
| Residual harm $H$ , LTT      | 2.8 [1.9, 4.1]   | –    | 84.8  | 15.0  |
| Residual harm $H$ , CRC      | 3.9 [2.8, 5.3]   | –    | 37.5  | 45.5  |

Three results, each contrary to what we pre-registered, follow from Table S4. First, certifying  $H$  at 5 % with LTT selects a threshold at which the vehicle fires in essentially every scenario (84.8 % unnecessary stops), and even then it exceeds the target in 15 % of resplits, above  $\delta$ , so it is not a valid certificate (N1 refuted). Second, the alert-level threshold’s residual harm on this fresh split is 5.0 % [3.7, 6.6], so its interval includes the 5 % target, unlike the 6.5 % on the main split (N2 refuted on fresh data; the interval is wide enough that a modest excess is not excluded). Third, CRC violates in 45.5 % of resplits, but LTT also exceeds  $\delta$ , so a non-monotone-aware calibrator is not by itself sufficient (N3 refuted). A joint certificate on residual harm and induced collisions over threshold and two MRM decelerations (2 and  $4 \text{ m/s}^2$ ) found no configuration other than always firing (N4 refuted). The practical reading is that, with this score and this maneuver, residual harm of 5 % or less is not reachable by tuning the trigger threshold alone at a useful operating point. A different maneuver or score is needed, not a different calibrator.

#### S6. SHIFT SEVERITY AND THE COST OF THE INTERVENTION (DESCRIPTIVE)

These two tables are computed from the stored records and test no hypothesis.

*Shift severity.* Table S5 measures how far each target is from the Argoverse 2 calibration set in the seven pre-deployment covariates used for the density ratio. The domain classifier is evaluated by 5-fold cross-validated AUROC (0.5 means indistinguishable), and the Argoverse 2 test split is a null control. nuScenes is much easier to tell apart than Waymo, has a Kish effective sample size of 66 against 288, and a 5.0 % floor of harm that no trigger can pre-empt. The certificate failed on the more distant target and held on the nearer one, which is consistent with shift severity mattering, but two targets cannot establish that, and the floor alone accounts for 5.0 of the 13.2 points missed on nuScenes.

TABLE S5

SHIFT SEVERITY AND THE ARGOVERSE 2-CERTIFIED THRESHOLD ( $\lambda = 1.47$ ). AUROC: CROSS-VALIDATED DOMAIN CLASSIFIER AGAINST THE ARGOVERSE 2 CALIBRATION SET. FLOOR: HARM PRESENT EVEN WHEN THE FALLBACK FIRES AT THE FIRST DECISION TICK. RATES IN %.

| Target             | $n$   | AUROC | $n_{\text{eff}}$ | Harm | Floor | Miss [CI]         |
|--------------------|-------|-------|------------------|------|-------|-------------------|
| AV2 test (control) | 1,996 | 0.50  | 1,978            | 15.1 | 0.0   | 3.7 [2.9, 4.6]    |
| Waymo              | 1,000 | 0.91  | 288              | 15.4 | 0.1   | 2.7 [1.9, 3.9]    |
| nuScenes           | 1,504 | 0.97  | 66               | 22.1 | 5.0   | 13.2 [11.6, 15.0] |

*Scenarios from the same scene are not independent.* The Wilson intervals in the main paper treat scenarios as independent. The nuScenes scenario identifiers contain the scene name, which allows a check: the 1,504 nuScenes scenarios in the main analysis come from only 37 scenes (41 per scene on average). Resampling whole scenes (cluster bootstrap, 10,000 draws) widens the interval for the miss rate of the Argoverse 2-certified threshold from [11.6, 15.0] to [6.1, 21.0], which still excludes the 5 % target, but only just. A second, largely disjoint set of nuScenes scenarios (1,342 scenarios from 23

scenes, one scene in common, collected for the replication analyses) gives a miss rate of 9.5 % with Wilson interval [8.1, 11.2] and cluster-bootstrap interval [3.9, 16.5], which includes the target. The first-tick floor also differs between the two samples (5.0 % against 0.0 %), so it varies strongly with which scenes are drawn. The direction of the failure therefore replicates, but with a few dozen scenes per sample the size of the nuScenes failure is far less certain than the main-paper intervals suggest. Argoverse 2 and Waymo identifiers are opaque, so the same check, and group-level resplits for the validity analysis, were not possible there.

*Where the induced collisions come from.* At the certified threshold on Argoverse 2 test, 2.6 % of scenarios [1.9, 3.3] suffer an induced collision. All 129 induced contact events are other vehicles striking the ego, 99 from behind and 30 from the side. They concentrate where the ego is slow or fires early. Scenarios in the two lower terciles of initial ego speed (the lowest starts at standstill, the middle one below 4.6 m/s) have induced rates of 3.6 % [2.4, 5.3] and 3.5 % [2.3, 5.1], against 0.6 % [0.2, 1.5] in the upper tercile, and scenarios in which the fallback fires at or before the median firing tick (tick 5, 2.5 s) have 9.2 % [6.8, 12.4] against 3.0 % [1.7, 5.1] for later firing. These contacts do not look like a follower caught off guard by hard braking. Of the 129, only 3.9 % occur within 2 s of the fallback firing, the median is 8.3 s after it, and 78 % come more than 5 s after it, by which time a vehicle braking at 4 m/s<sup>2</sup> from city speed has long been stationary. The 129 contacts come from only 51 scenarios, and 67 % of them are in scenarios where the ego starts at standstill, so the fallback there changes little about what the ego does. We therefore cannot attribute this cost to the act of stopping. It may instead come from simulated traffic, driven along logged paths with a car-following model, that does not yield to a vehicle which stays stationary in its path for many seconds, which a real driver would also be expected to avoid (released with the code). The 2.6 % induced rate should be read as an upper bound on the cost that stopping itself causes in this simulator.

To grade severity we re-ran the single forced rollout of each of the 51 scenarios with instrumentation (Amendment 9, descriptive). The re-run reproduces the stored induced-contact steps exactly in 34 scenarios and the first induced contact in 49, but later contacts differ in 17, so the contact dynamics after a first impact are not run-to-run reproducible, which is one more way in which the forced-rollout table is exact only up to the first contact. At the first induced contact the ego's speed has a median of 0.6 m/s (27 % below 0.1 m/s and 65 % below 1 m/s), the other vehicle's speed a median of 3.6 m/s, and the speed difference a median of 2.4 m/s (quartiles 1.2 and 5.8, maximum 11.8; this is a scalar speed difference, not a closing speed along the contact axis). Nineteen of the 51 first contacts are lateral and 32 are from behind, a median 6.5 s after firing. At the firing step an agent was in the ego's lane behind it in 39 of the 51 scenarios, at a median gap of 8.3 m and a median speed of 0 m/s. These are therefore low- to moderate-speed contacts with a slow or stationary ego by traffic that keeps moving along its logged path, not hard-braking surprises. The fallback also costs progress: where it fires, the vehicle stops for 7.4 s on average and completes 49 % of the route, against

94 % where it does not. The rates count scenarios, not contacts.

## S7. DIRECT EXECUTION AGAINST THE LOOKUP

The main paper evaluates every trigger by lookup in a table of forced-fire rollouts, which is exact when a run under the trigger coincides with the reference run until it fires (Observation 1). The assumption is measured in the forced runs but was not tested by executing a trigger. This check, pre-registered as Amendment 7 and extended in Amendment 9, does so. It draws two independent random samples of stored Argoverse 2 test scenarios (120 and 140 scenarios, 252 distinct after merging by seed) and runs four geometric triggers live, with the predictor scoring at every decision tick and no score replay or forced-fire table: the certified LTT threshold with zero and with one decision tick of latency, and the thresholds at the 25th and 75th percentile of the calibration peak scores (about 75 % and 25 % of scenarios fire). Each run is scored with the same nuPlan harm definition and compared, scenario by scenario, with the lookup at the same threshold and latency. No direct run failed. (The first draw was first analyzed as a snapshot of 61 scenarios, because the stop signal described in Amendment 7 did not take effect and the job ran to completion. The figures here supersede that snapshot.)

TABLE S6  
DIRECT EXECUTION VERSUS LOOKUP, 252 SCENARIOS  $\times$  4 TRIGGERS = 1,008 PAIRS. DISAGREEMENT: PAIRS IN WHICH THE TWO DIFFER.

| Quantity                   | Disagree  | Wilson 95 % upper bound |
|----------------------------|-----------|-------------------------|
| Firing step (or no firing) | 0 / 1,008 | 0.4 %                   |
| Missed intervention        | 0 / 1,008 | 0.4 %                   |
| Unnecessary stop           | 0 / 1,008 | 0.4 %                   |
| Induced collision          | 0 / 1,008 | 0.4 %                   |
| Harm anywhere in the run   | 1 / 1,008 | 0.6 %                   |

Table S6 shows no disagreement in any alert-level quantity or in the induced collisions. The aggregate rates are identical for every trigger (for example 3.6 % missed and 27.8 % unnecessary stops for the certified threshold, and 4.8 % missed with one tick of latency). One pair differs in harm anywhere in the run (seed 1934, certified threshold): the lookup records harm after the fallback and the direct run does not. We did not trace its cause. The re-run of induced collisions in Section S6 shows that contact events after a first impact are not always reproducible from run to run, which is a plausible source. Route completion is identical in 454 of the 1,008 pairs and differs by 0.0002 on average and 0.036 at most. The pre-registered rule, that the lookup is refuted as a substitute if any quantity disagrees in more than 5 % of pairs, is therefore not triggered, and the upper bound on the disagreement rate of the alert-level quantities is 0.4 %. The check covers one dataset, one predictor and one maneuver, and does not test stochastic traffic, for which assumption (A1) would not hold and the lookup would estimate the outcome under a different random draw than the one the trigger would have met. The magnitudes of the pre-fire deviations in the forced runs were not stored, only whether they exceeded 1 cm.

## S8. REPLICATION ON FRESH DATA AND A SECOND PREDICTOR

Table S7 gives the numbers behind the replication analysis in the main paper. Violation frequencies over 200 resplits have Wilson 95 % intervals of [4.6, 12.0] for LTT with the primary predictor and [5.8, 13.8] with the second, so, as in the main split, the point estimates meet  $\delta = 0.10$  but their Monte Carlo intervals do not exclude a violation frequency above it. The uncertified baselines (46–51.5 %) are far outside it. The gap between the geometric and the confidence trigger also replicates: with the same target, LTT stops 29.0 % [26.1, 32.0] of scenarios against 67.0 % [63.9, 70.0] for the tuned confidence trigger on the fresh split, and 30.6 % [27.7, 33.7] against 68.3 % [65.1, 71.2] with the second predictor.

TABLE S7

REPLICATION ON FRESH, DISJOINT DATA ( $n_{\text{cal}} = n_{\text{test}} = 904$ ).  
VIOLATION: FRACTION OF 200 RESPLITS WITH TEST MISS  $> \alpha$ .

| Method          | Fresh held-out |           | 2nd predictor |           |
|-----------------|----------------|-----------|---------------|-----------|
|                 | Miss (%)       | Viol. (%) | Miss (%)      | Viol. (%) |
| LTT (geometric) | 3.4            | 7.5       | 3.9           | 9.0       |
| CRC (geometric) | 4.3            | 46.0      | 4.6           | 47.5      |
| Tuned geometric | 4.3            | 51.5      | 4.6           | 51.0      |
| T1 confidence   | 4.8            | 50.0      | 4.6           | 46.0      |

*Ensemble baseline with the second predictor.* The replication above lacked the Monte Carlo dropout trigger. We re-ran only the reference run of every fresh calibration and test scenario with five dropout passes (Amendment 9), reusing the forced-fire outcomes, which do not depend on any score. The re-run reproduced the stored reference (tick count and geometric trace) in 865 of 904 calibration and 867 of 904 test scenarios, and we dropped the others (4.3 % and 4.1 %), so these figures use 865 and 867 scenarios. On the test scenarios the certified geometric trigger misses 3.2 % [2.2, 4.6] with 26.4 % unnecessary stops, the tuned confidence trigger 5.0 % [3.7, 6.6] with 66.2 %, and the tuned ensemble trigger 5.7 % [4.3, 7.4] with 59.6 %. The tuned ensemble trigger therefore misses the 5 % target, which the pre-registered comparison “among triggers that meet the target” did not anticipate, so the rule as written cannot be evaluated for it. Unconditionally, the certified trigger stops 33.2 points less than the ensemble trigger (paired 95 % CI  $[-37.5, -29.0]$ ) and 39.8 points less than the confidence trigger ( $[-43.7, -35.8]$ ), with a lower miss rate than both. The geometric-versus-ensemble gap is thus reproduced with a second predictor, and the open-loop conformal comparison alone remains untested there.

## S9. CERTIFYING UNDER THE DEPLOYED LATENCY

The main paper shows that a certificate calibrated at zero latency fails when the actuation chain is slower. Because the loss at latency  $\ell$  decision ticks is the stored miss indicator at tick  $\tau(X) + \ell$ , LTT can be run directly on that loss without new simulation (Amendment 8, pre-registered before the run). We certify on the ArgoVerse 2 calibration set at  $\ell \in \{0, 1, 2\}$  ticks (0, 0.5, 1.0 s), deploy at the same latency on the test set, and repeat over the same 200 half-and-half resplits as the main paper.

TABLE S8

LTT CALIBRATED AT THE DEPLOYED LATENCY (“AWARE”) VERSUS THE ZERO-LATENCY CERTIFICATE DEPLOYED AT LATENCY  $\ell$  (“NAIVE”), ARGOVERSE 2 TEST. MISS AND STOPS ON THE TEST SET, RATES IN %. VIOL.: FRACTION OF 200 RESPLITS WITH TEST MISS ABOVE 5 %, WITH WILSON 95 % INTERVALS.

| $\ell$ | Naive miss | Naive viol.       | Aware miss | Aware stops | Aware viol.      |
|--------|------------|-------------------|------------|-------------|------------------|
| 0      | 3.7        | 9.5 [6.2, 14.4]   | 3.7        | 28.5        | 9.5 [6.2, 14.4]  |
| 1      | 5.3        | 92.0 [87.4, 95.0] | 3.6        | 34.4        | 12.0 [8.2, 17.2] |
| 2      | 9.1        | 100 [98.1, 100]   | 4.1        | 42.5        | 11.5 [7.8, 16.7] |

The zero-latency certificate is not valid under latency, as claimed. The latency-aware certificate brings the miss rate back below 5 % on the test set, at a cost of 6 and 14 points of unnecessary stops. Its violation frequency, however, is 12.0 % and 11.5 % at one and two ticks, above the  $\delta = 0.10$  we pre-registered, so by our own rule the claim that latency-aware certification restores validity was refuted under the uncorrected count. Under the corrected count of Amendment 11 (Section S17), 1.5 % and 0.5 %, the certificate is indeterminate, not violated, and the pre-registered re-reading holds. Three readings are possible and we cannot separate them. The Wilson intervals include 0.10. The statistic counts resplits whose finite test half exceeds 5 %, not resplits whose population risk does, and the zero-latency case is already 9.5 %. And the loss at a positive latency is less smooth in  $\lambda$ , which may weaken the fixed-sequence test in practice. The guarantee itself is unchanged, since the loss is still a fixed function of the scenario and the threshold, but we report the frequency as measured. Across resplits the mean unnecessary-stop rate of the latency-aware certificate is 27.6, 31.8 and 57.0 %, so at two ticks the cost depends strongly on the calibration draw.

## S10. NATURAL SHIFTS INSIDE ARGOVERSE 2: LEAVE-ONE-CITY-OUT

The main paper’s shift evidence rests on two target datasets. ArgoVerse 2 labels each scenario with one of six cities, which gives six further natural shifts at no simulation cost. For each city we run LTT ( $\alpha = 0.05$ ,  $\delta = 0.10$ , geometric score) on all stored calibration and test scenarios outside that city (about 3,000 to 3,800 scenarios) and deploy the threshold on the held-out city. We pre-registered (Amendment 9) that the certificate would count as transferring if the held-out miss rate were at most 5 % in at least 80 % of the cities.

TABLE S9

LEAVE-ONE-CITY-OUT ON ARGOVERSE 2 (POOLED CALIBRATION AND TEST SCENARIOS,  $n = 3,990$ ). HARM: HARM RATE OF THE HELD-OUT CITY. MISS AND STOPS: HELD-OUT CITY AT THE THRESHOLD CERTIFIED ON THE OTHER CITIES, WITH WILSON 95 % INTERVALS FOR THE MISS RATE. TUNED: MISS OF THE UNCERTIFIED THRESHOLD TUNED ON THE OTHER CITIES. RATES IN %.

| Held-out city  | $n$ | Harm | LTT miss [CI]  | LTT stops | Tuned miss |
|----------------|-----|------|----------------|-----------|------------|
| Dearborn       | 546 | 8.4  | 2.0 [1.1, 3.6] | 17.0      | 2.9        |
| Palo Alto      | 229 | 12.7 | 4.4 [2.4, 7.9] | 21.0      | 4.8        |
| Austin         | 931 | 14.2 | 3.7 [2.6, 5.1] | 26.7      | 4.1        |
| Pittsburgh     | 870 | 15.4 | 4.4 [3.2, 5.9] | 30.7      | 5.1        |
| Washington, DC | 450 | 19.3 | 5.3 [3.6, 7.8] | 26.7      | 6.2        |
| Miami          | 964 | 19.7 | 6.1 [4.8, 7.8] | 30.2      | 7.0        |

The certificate met the target in four of the six cities (67 %), below the 80 % we required, so the pre-registered claim is refuted. The uncertified threshold met it in three. The two failures, Miami and Washington, DC, are the two cities with the highest harm rates, and in only Miami does the interval exclude 5 %. The failure is not an artifact of holding the city out: the threshold certified on all cities, which saw them, also misses 5.2 % and 5.1 % there. This is what a *marginal* guarantee implies, since the miss rate is the harm rate times the fraction of harmful scenarios left unalerted. Post hoc and descriptive, across these six cities and the two datasets of the main paper, the held-out miss rate and the harm rate of the target have a Spearman correlation of 0.90 ( $n = 8$ , so this is suggestive only). A shift in the base rate of harm, which an unlabeled-covariate reweighting cannot see, is therefore a plausible common cause of the failures, and it would also explain why the certificate failed on nuScenes (22.1 % harm) but held on Waymo (15.4 %).

### S11. SHIFT IN THE TRAFFIC BEHAVIOR MODEL

All main results use reactive IDM traffic. To test a different behavior model we deploy the thresholds certified on reactive calibration data on the non-reactive log-replay arm, paired by scenario seed with the reactive test arm (378 scenarios with complete records). The harm base rate is 16.7 % under reactive traffic and 17.7 % under replay. The certified geometric trigger misses 3.7 % [2.2, 6.1] on replay traffic, so the pre-registered claim that it still meets the 5 % target is not refuted. The tuned confidence trigger also meets the target (3.4 %), but at 71.7 % unnecessary stops against 35.4 % for the certified geometric trigger (paired difference  $-36.2$  points, 95 % CI  $[-41.8, -30.7]$ ). The stop-rate advantage therefore persists under a second traffic model, although both are rule-based and neither is a learned agent model.

### S12. SPATIAL-GROUP DEPENDENCE

Scenarios from the same place or drive may be dependent, which would weaken the exchangeability that the guarantee needs. The converted data carry no recording identifier for Argoverse 2 and Waymo, so (Amendment 10, pre-registered) we group scenarios by the ego’s initial position: single-linkage components of scenarios whose ego start points lie within 150 m of each other, formed separately for each Argoverse 2 city. This is a proxy, not a recording identifier. It gives 547 groups over the 3,990 pooled Argoverse 2 scenarios (the largest has 649, 16 % of them, the median group is a single scenario) and 664 groups over the 1,000 Waymo scenarios (the largest has 13).

We repeat the main validity analysis with whole groups assigned to calibration or test (200 resplits, calibration share about 53 %). The violation frequency of LTT is 17.0 % (Wilson 95 % interval [12.4, 22.8]) against 9.5 % for random resplits, and the mean test miss rate is 4.2 %. The pre-registered claim that it stays at or below  $\delta = 0.10$  is therefore refuted under the uncorrected count (it is indeterminate, at 1.0 %, under the corrected count of Section S17). A post hoc sensitivity at 50 m, which gives 1,796 groups with the largest at 79

scenarios, yields 14.0 % [9.9, 19.5]. Dependence between nearby scenarios is thus one more reason the high-probability statement is weaker in practice than in theory, although random and group-wise splits differ by less than the Monte Carlo error of either would suggest at 200 resplits.

The miss-rate intervals are far less sensitive. A group bootstrap of the certified threshold’s miss rate gives [3.0, 4.5] % on the Argoverse 2 test scenarios (360 groups; Wilson [2.9, 4.6]) and [1.7, 3.8] % on Waymo (664 groups; Wilson [1.9, 3.9]). Place-level dependence therefore does not change the point conclusions about the miss rate, including the conservative result on Waymo.

### S13. A SECOND ARCHITECTURE: WAYFORMER

Every main result uses AutoBot, and the second predictor of Section S8 is also an AutoBot. To test whether the findings are specific to the architecture, we repeated the full closed-loop campaign (Amendment 10, pre-registered) on the same 904 calibration and 904 test scenarios with a Wayformer forecaster from the same framework ( $\text{minADE}_6 = 0.967$ , a partly trained checkpoint, the same 2.1 s history and 6 s horizon). A check on identical simulator states found a median 3 s endpoint disagreement of 1.6 m with AutoBot, so the wrapper produces comparable forecasts. T2 and T3 are not repeated.

TABLE S10  
WAYFORMER ARM, ARGOVERSE 2 FRESH TEST SCENARIOS ( $n = 904$ ).  
VIOL.: FRACTION OF 200 RESPLITS WITH TEST MISS ABOVE 5 %. RATES IN %.

| Method              | Miss [95 % CI] | Unnec. stops | Viol. |
|---------------------|----------------|--------------|-------|
| LTT (geometric)     | 3.1 [2.2, 4.4] | 41.8         | 7.0   |
| Tuned geometric     | 3.8 [2.7, 5.2] | 36.7         | 48.0  |
| CRC (geometric)     | —              | —            | 44.0  |
| Tuned T1 confidence | 4.2 [3.1, 5.7] | 67.8         | 53.0  |

Both parts of the pre-registered claim hold. LTT is valid (7.0 % violations, Wilson interval [4.2, 11.4], and a test miss of 3.1 %), and the certified geometric trigger stops 26.0 points less than the tuned confidence trigger (paired 95 % CI  $[-29.8, -22.1]$ ), which meets the target with 4.2 % missed. The uncertified geometric threshold and CRC violate the target in 48 % and 44 % of resplits, as with AutoBot. The certificate itself and the geometric advantage therefore do not depend on the architecture. The price does depend on the predictor: with Wayformer the certified trigger stops 41.8 % of scenarios, against 26 to 29 % with the two AutoBot models, which is consistent with a less accurate forecaster needing a more conservative threshold to meet the same target.

### S14. PHYSICS BASELINES, ABLATIONS AND OPEN-LOOP RANKING

The main paper’s Table I adds a time-to-collision (TTC) trigger, certified with the same procedure. Because the harm label is itself defined by time-to-collision, the TTC trigger is close to the label, so we repeat the comparison under the alternative harm definitions of Section V (post hoc, not pre-registered). The certified TTC trigger needs fewer unnecessary stops than the predictor-based geometric trigger under three of

the four definitions: collision only ( $\alpha = 0.01$ ) 18.8 % against 45.1 %, TTC < 0.5 s 1.6 % against 15.8 %, the headline TTC < 0.95 s 7.1 % against 28.5 %. Under TTC < 1.5 s the certified TTC threshold is very conservative (77.5 % against 34.0 %). A headway-distance trigger is worse than the geometric trigger under all four. The ablations below and the open-loop ranking behind Section V-B follow.

TABLE S11

ABLATIONS ON THE ARGOVERSE 2 TEST SET, CERTIFIED BY LTT.

| Variant                             | Miss (%)   | Unnec. stops (%) |
|-------------------------------------|------------|------------------|
| Full score, LTT, no latency         | 3.7        | 28.5             |
| Route-following ego corridor        | 0.0        | 84.9             |
| Width-only isotropic radius         | 3.6        | 27.5             |
| Confidence instead of geometry      | 3.6        | 75.8             |
| Ensemble spread instead of geometry | 3.6        | 72.3             |
| CRC instead of LTT                  | 4.3        | 26.0             |
| Empirical tuning instead of LTT     | 4.3        | 26.0             |
| +1 decision tick latency (0.5 s)    | <b>5.3</b> | 26.3             |
| +2 decision ticks latency (1.0 s)   | <b>9.1</b> | 23.4             |

TABLE S12

OPEN-LOOP AUROC VERSUS CLOSED-LOOP UNNECESSARY STOPS AT THE CERTIFIED THRESHOLD, ARGOVERSE 2 TEST.

| Score                       | AUROC | Unnec. stops (%) |
|-----------------------------|-------|------------------|
| Geometric, isotropic radius | 0.843 | 27.5             |
| Geometric, length-aware box | 0.832 | 28.5             |
| Predictor gap               | 0.817 | 31.8             |
| Geometric, route corridor   | 0.577 | 84.9             |
| Confidence                  | 0.448 | 75.8             |
| Ensemble spread             | 0.442 | 72.3             |

TABLE S13

LTT (GEOMETRIC) VERSUS THE BETTER OF T1/T2 UNDER ALTERNATIVE HARM DEFINITIONS, ARGOVERSE 2 TEST. UNNECESSARY-STOP RATES IN %; “BEST” IS THE BETTER OF T1/T2 AMONG THOSE MEETING THE MISS TARGET, AND THE PAIRED 95 % CI IS OF THE STOP-RATE DIFFERENCE.

| Harm definition               | LTT  | Best | Which | Diff. [95 % CI]      |
|-------------------------------|------|------|-------|----------------------|
| Collision only, $\alpha=0.01$ | 45.1 | 83.9 | T2    | −38.8 [−41.4, −36.2] |
| TTC < 0.5 s                   | 15.8 | 38.6 | T1    | −22.7 [−25.5, −20.1] |
| TTC < 0.95 s (headline)       | 28.5 | 68.3 | T1    | −39.8 [−42.4, −37.2] |
| TTC < 1.5 s                   | 34.0 | 68.3 | T2    | −34.4 [−36.8, −31.9] |

### S15. COUNTERFACTUAL TRIGGER: SAFE WINDOWS, SHIFT AND PREVALENCE CORRECTION

*Safe windows.* On the 1,994 training scenarios, 15.8 % have harm in the no-fallback run. Harm is unavoidable (empty safe window) in 0.35 % of scenarios, so almost all harm is preventable by some firing tick. In 12.7 % of scenarios the window is not contiguous (a safe tick, an unsafe tick, then a safe tick again), which is why the outcome loss is certified directly instead of through a union bound on too-late and too-early firing.

*Prevalence-corrected certificate (refuted).* A marginal miss rate is the harm rate times the miss rate among harmful scenarios, and the held-out miss rate tracked the target’s harm rate. We therefore estimated each target’s harm rate from

unlabeled data with black-box shift estimation (the pilot alert as classifier) and certified a class-conditional miss rate at  $\alpha$  over the estimate’s upper bound (Amendment 11, N-A). The estimate was unreliable (Table S14): the alert rate shifts for reasons other than harm rate, so the label-shift assumption fails. The corrected certificate met the marginal target on 5 of 8 targets, against 6 of 8 for plain LTT, and the pre-registered claim (at least 7 of 8 and more than plain LTT) is refuted.

TABLE S14

BLACK-BOX SHIFT ESTIMATE OF THE TARGET HARM RATE AND MARGINAL MISS AT THE TARGET, UNWEIGHTED LTT VERSUS THE PREVALENCE-CORRECTED CERTIFICATE (PC). RATES IN %.

| Target         | True harm | Estimate | Upper bound | LTT miss | PC miss |
|----------------|-----------|----------|-------------|----------|---------|
| Austin         | 14.7      | 8.7      | 13.4        | 4.1      | 4.9     |
| Dearborn       | 8.4       | 0.0      | 0.0         | 2.2      | 8.4     |
| Miami          | 20.0      | 33.9     | 38.4        | 6.1      | 1.9     |
| Palo Alto      | 11.6      | 0.0      | 6.5         | 3.9      | 7.7     |
| Pittsburgh     | 15.5      | 25.1     | 29.5        | 4.6      | 2.0     |
| Washington, DC | 17.6      | 21.8     | 28.1        | 4.2      | 2.1     |
| nuScenes       | 22.1      | 4.8      | 9.4         | 14.0     | 16.3    |
| Waymo          | 15.4      | 46.3     | 51.7        | 2.8      | 0.5     |

*Counterfactual trigger under shift (refuted).* Applying the same correction to the outcome target of the learned trigger (tightening 7.5 % by the ratio of the source harm rate to the upper bound) made it meet the target on 4 of 8 targets and refuse to certify on 3, against 6 of 8 without the correction (fails without it: Miami 10.8 %, nuScenes 8.2 %). The pre-registered claim that the correction helps is refuted, for the same reason.

### S16. OUTCOME AUDIT WITH AN UPPER CONFIDENCE BOUND

The alert-level threshold (Learn-then-Test on the Argoverse 2 calibration split, 5 % target) was applied to a fresh split of 904 scenarios, where its residual harm is 6.7 % and total harm (with induced collisions) 8.1 %. One-sided 90 % Clopper–Pearson upper bounds are 7.9 % and 9.4 %. The realized harm on each target is compared with the bound in Table S15. The bound covers the realized residual harm on 7 of 9 targets (78 %), below the 90 % we required, so the pre-registered claim is refuted. The two misses are nuScenes, a real shift, and Palo Alto, where 8.0 % against 7.9 % on 253 scenarios is within sampling noise. The audit is valid when the target resembles the audit split and must be recomputed on target-like labeled data otherwise.

TABLE S15

REALIZED RESIDUAL HARM  $O$  AND TOTAL HARM  $O_{\text{tot}}$  AGAINST THE AUDIT UPPER BOUNDS (7.9 % AND 9.4 %). RATES IN %.

| Target                       | $O$  | $O_{\text{tot}}$ | Covered |
|------------------------------|------|------------------|---------|
| Fresh test (in distribution) | 4.8  | 7.2              | yes     |
| Austin                       | 7.0  | 8.6              | yes     |
| Dearborn                     | 4.4  | 5.1              | yes     |
| Miami                        | 6.8  | 8.3              | yes     |
| Palo Alto                    | 8.0  | 10.0             | no      |
| Pittsburgh                   | 6.3  | 8.4              | yes     |
| Washington, DC               | 6.7  | 8.1              | yes     |
| nuScenes                     | 15.6 | 16.4             | no      |
| Waymo                        | 4.3  | 8.3              | yes     |

### S17. CORRECTED VALIDITY AND THE HYPOTHESIS LEDGER

The violation frequency reported in the main paper counts resplits whose test-half miss exceeds  $\alpha$ , which also counts test-sample noise. We also report a corrected count that only counts resplits whose one-sided 95 % Wilson lower bound exceeds  $\alpha$ , and call a method *valid* if the uncorrected count is at most  $\delta$ , *indeterminate* if only the corrected count is, and *violated* otherwise (Amendment 11). Table S16 gives every method. The separation of Learn-then-Test from the uncertified methods survives the correction, whereas the near-boundary cases (latency-aware calibration, group-wise resplits) become indeterminate rather than violated.

TABLE S16  
VIOLATION FREQUENCY OVER 200 RESPLITS, UNCORRECTED /  
CORRECTED, AND VERDICT. RATES IN %.

| Method                      | Unc. / corr. | Verdict       |
|-----------------------------|--------------|---------------|
| LTT (geometric)             | 9.5 / 0.0    | valid         |
| Tuned geometric             | 43.5 / 15.0  | violated      |
| CRC                         | 41.0 / 12.0  | violated      |
| Tuned T1 confidence         | 50.5 / 14.0  | violated      |
| LTT latency-aware, 0.5 s    | 12.0 / 1.5   | indeterminate |
| LTT latency-aware, 1.0 s    | 11.5 / 0.5   | indeterminate |
| Zero-latency LTT at 0.5 s   | 92.0 / 53.0  | violated      |
| Zero-latency LTT at 1.0 s   | 100 / 100    | violated      |
| LTT, fresh data             | 7.5 / 0.5    | valid         |
| LTT, second predictor       | 9.0 / 0.0    | valid         |
| LTT, Wayformer              | 7.0 / 0.0    | valid         |
| LTT, spatial groups (150 m) | 17.0 / 1.0   | indeterminate |
| TTC, LTT                    | 1.0 / 1.0    | valid         |
| CDT (100 resplits)          | 59.0 / 0.0   | indeterminate |
| ACI (100 resplits)          | 0.0 / 0.0    | valid         |

Every claim of Amendment 11 and its addendum was fixed before it was run, and the ledger records each verdict.

TABLE S17  
AMENDMENT 11 CLAIMS AND VERDICTS.

| Claim                                                                                                | Verdict |
|------------------------------------------------------------------------------------------------------|---------|
| P1: the predictor-based trigger stops less than a refuted physics TTC trigger                        |         |
| N-A: prevalence-corrected certificate meets the target on at least 7 of 8 shifts and beats plain LTT | refuted |
| N-B: latency-aware LTT is not violated at 0.5 and 1.0 s                                              | holds   |
| N-C: the outcome audit bound covers at least 90 % of targets                                         | refuted |
| CSWC (i): outcome certificate valid at 7.5 %                                                         | holds   |
| CSWC (ii): learned trigger stops less than geometric and TTC at zero latency                         | refuted |
| CSWC (iii): prevalence correction helps the learned trigger under shift                              | refuted |
| 11b: learned trigger valid and cheapest at 0.5 and 1.0 s                                             | holds   |

### S18. OUTCOME PARTITION OF THE FALLBACK

The table partitions every scenario by what happens after the fallback fires, as described in the main paper.

TABLE S18  
OUTCOME PARTITION (% OF SCENARIOS). RESIDUAL = MISS + TOO LATE  
+ NEW HARM. INDUCED: NOT-AT-FAULT COLLISION THE NO-FALLBACK  
RUN AVOIDS.

| Data     | Trigger         | Miss | Too late | Prev. | New harm | Resid. / Induced |
|----------|-----------------|------|----------|-------|----------|------------------|
| AV2      | No fallback     | 15.1 | 0.0      | 0.0   | 0.0      | 15.1 / 0.0       |
| AV2      | LTT (certified) | 3.7  | 1.7      | 9.8   | 1.1      | 6.5 / 2.6        |
| AV2      | Tuned geometric | 4.3  | 1.7      | 9.1   | 0.9      | 6.9 / 2.3        |
| AV2      | Always fire     | 0.0  | 1.0      | 14.1  | 3.1      | 4.1 / 6.5        |
| nuScenes | LTT (AV2-cert.) | 13.2 | 1.0      | 8.2   | 1.7      | 15.6 / 2.4       |
| nuScenes | Always fire     | 5.0  | 1.6      | 16.2  | 3.8      | 9.7 / 6.4        |